

Supporting Information for: Suppression of Pure Dephasing and Enhancement of Excitonic Coherence via Non-Markovian Spectral Bath Engineering

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I. FMO HAMILTONIAN AND SPECTRAL DENSITY PARAMETERIZATION

The analytical details of the Process Tensor Hierarchy of Pure States (PT-HOPS), the Padé Matsubara decompositions, and the full 12-test numerical validation benchmarks isolating numerical positivity and Hierarchical Equations of Motion (HEOM) deviations are strictly documented inside the primary manuscript text.

This Supporting Information supplements those analytical proofs by explicitly reproducing the exact 7-site bacteriochlorophyll-*a* matrices (obtained from Adolphs and Renger[?]) utilized within our systemic calculations to assist independent computational replication vectors.

A. Site Energies

TABLE I. FMO Site Energies (cm⁻¹)

ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7
12400	12550	12250	12350	12450	12600	12500

B. Electronic Couplings

The inter-chromophore electronic coupling matrix J_{nm} (in cm⁻¹):

$$\mathbf{J} = \begin{pmatrix} 0 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 0 & 30.8 & 8.2 & 0.7 & -11.9 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 0 & 1.2 & -6.3 & 1.7 \\ 6.7 & 0.7 & -2.2 & 1.2 & 0 & 9.6 & -6.0 \\ -13.7 & -11.9 & -9.6 & -6.3 & 9.6 & 0 & 89.7 \\ -9.9 & 4.3 & 6.0 & 1.7 & -6.0 & 89.7 & 0 \end{pmatrix}. \quad (1)$$

C. Spectral Density Parameters

The underlying continuous Drude-Lorentz environment is mapped alongside four discrete vibronic bounds to precisely model non-Markovian oscillations.

TABLE II. Systematic Spectral Density Operating Parameters

Parameter Constraint	Value Bound
Drude-Lorentz reorganization (λ_{DL})	35 cm ⁻¹
Drude-Lorentz thermal cutoff (γ_{DL})	50 cm ⁻¹
Underdamped Vibronic mode 1 (ω_1)	150 cm ⁻¹
Underdamped Vibronic mode 2 (ω_2)	200 cm ⁻¹
Underdamped Vibronic mode 3 (ω_3)	575 cm ⁻¹
Underdamped Vibronic mode 4 (ω_4)	1 185 cm ⁻¹
Huang-Rhys coefficient S_1	0.05
Huang-Rhys coefficient S_2	0.02
Huang-Rhys coefficient S_3	0.01
Huang-Rhys coefficient S_4	0.005

¹Adolphs, J.; Renger, T. Biophys. J. **2006**, *91*, 2778–2797.